

Frontier Is Not Fate

What does a better AI model actually change?

Follow it into a workflow. Then ask what else has to happen before the improvement becomes a useful outcome.

One supported starting point: in the ECB's Q4 2025 survey, **33%** of responding firms reported very infrequent or experimental AI use; **7%** reported significant use.

Self-reported intensity among weighted SAFE respondents. This tells us how firms report using AI; it does not measure a productivity gain. [Source and limits](#)

Capability and complements can both matter. Their effects depend on the task, quality threshold, workflow and beneficiary. This study traces those links; it does not estimate a national return to AI.

What has actually changed in work?

Start by separating reported use from measured work. A firm can tick "yes" while still experimenting. Surveys can tell us about reach, intensity or breadth; none of those answers alone establishes a completed outcome.

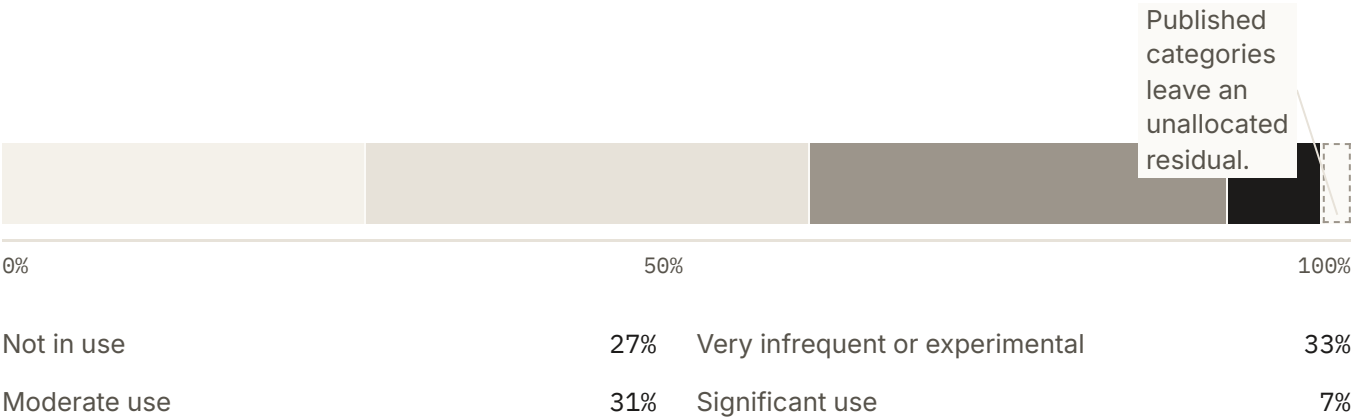
Observed: reported use has several meanings.

Percent of firms within each source's own universe. ECB intensity, Census function breadth and Eurostat firm-size adoption stay separate.

Each panel retains its own axis, denominator, and survey universe. Values are source-specific and must not be pooled or ranked.

Most reported use is experimental or moderate.

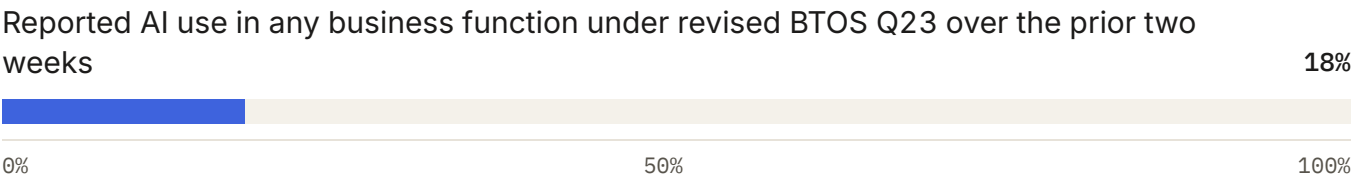
AXIS 0-100% · WEIGHTED SAFE RESPONDENTS



ECB SAFE Q4 2025 – source and caveat · denominator: All firms responding to SAFE QA1, weighted

Reach and functional breadth answer different questions.

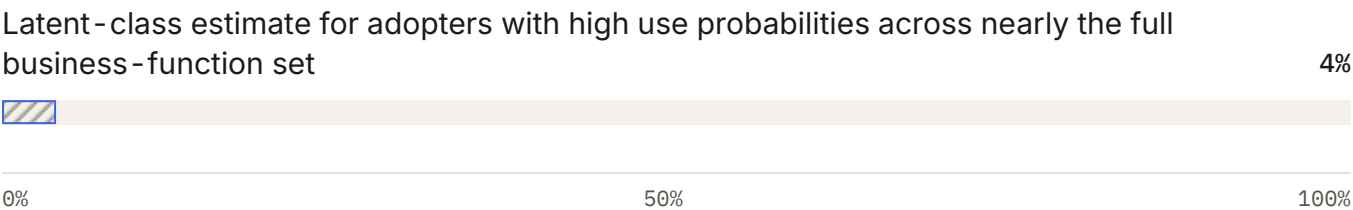
ALL EMPLOYER BUSINESSES · AXIS 0-100%



Q24 FUNCTIONAL USERS ONLY · AXIS 0-100%



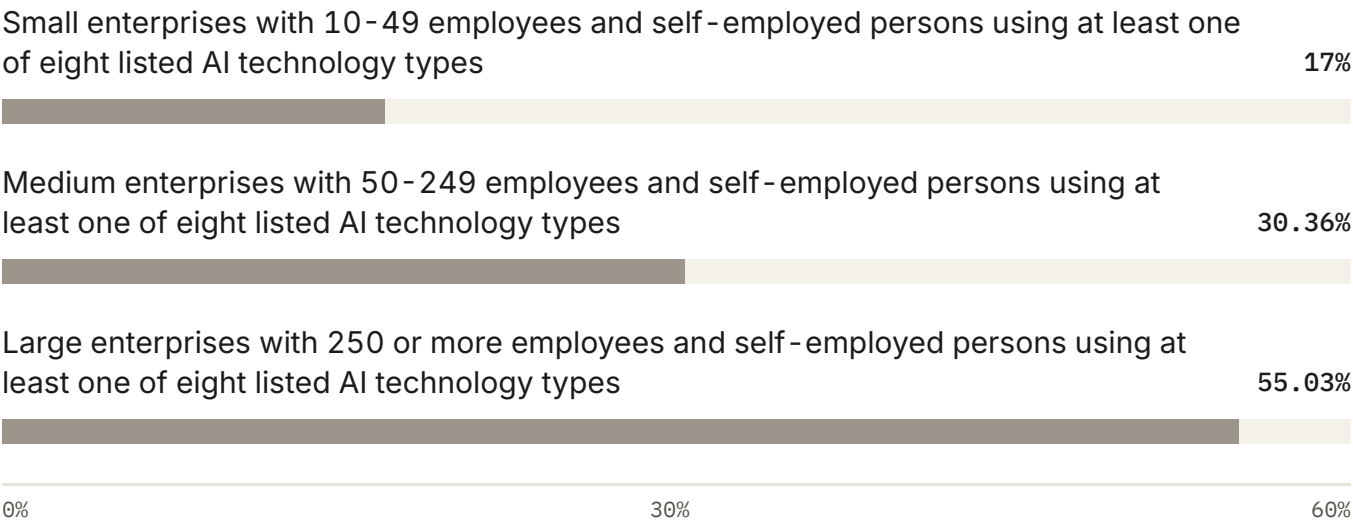
MODEL ESTIMATE



Census CES Working Paper 26-25 – source and caveat · Q23 denominator: All in-scope employer businesses responding to Q23, firm-weighted · Q24 denominator: Firms reporting AI use in at least one of 15 Q24 business functions, firm-weighted

Adoption is uneven across firm size.

WITHIN EACH SIZE UNIVERSE · AXIS 0-60%



Diffusion context only. This panel does not measure depth of use.

Eurostat 2025 – source and caveat · universe: Enterprises with 10 or more employees and self-employed persons in covered NACE Rev. 2 sectors; micro-enterprises are optional and outside the harmonized core

A different firm universe gives context only.

The NBS Fifth National Economic Census covers a different enterprise universe. It cannot supply a comparable all-firm adoption rate or a national ranking here.

The contextual observation is retained in the research record, unplotted. Load-bearing translated wording awaits native-language human review.

NBS – scope and translation restriction

ECB SAFE Q4 2025; Census CES Working Paper 26-25; Eurostat 2025 enterprise AI use. NBS context retained, unplotted. **OBSERVED** DEFINITIONS DIFFER

Evidence and caveats

All firms responding to SAFE QA1, weighted; 5,067-firm Q4 ECB-round sample across 12 euro-area countries with sector exclusions. The four whole-number categories total 98%; the residual may reflect don't-know responses and rounding, so do not derive 73% any use.

Denominator is all firms responding to SAFE QA1, weighted, in the 5,067-firm Q4 ECB-round private-sector sample across 12 euro-area countries; eight smallest euro-area countries and documented sectors are excluded. This is a source-specific category interpretation; do not infer a harmonized deep-use threshold or calculate a 73% any-use share from the 27% no-use value.

Firm-weighted denominator is businesses responding to Q23 within the nonfarm employer-business BTOS universe and documented exclusions. The paper headline rounds to 18%; Table C.7 reports 17.9%. Revised wording began 2025-11-17.

Denominator is firms reporting AI use in at least one Q24 function within the nonfarm U.S. employer-business universe covering single- and multi-location businesses in the United States, District of Columbia, and Puerto Rico, subject to documented industry exclusions; it is not the Q23 prior-two-week adopter group. The 4% is model-derived, does not mean use in every function, and does not form a partition with 57%.

The study universe is nonfarm U.S. employer businesses, single- and multi-location, in the United States, District of Columbia, and Puerto Rico, subject to documented BTOS industry exclusions; the analysis uses supplement respondents. The paper states that regressions have no causal interpretation and that CES papers do not imply Census Bureau endorsement.

Denominators are enterprises with 10+ employees and self-employed persons in covered NACE activities, with size-specific subgroups; micro-enterprises are outside the harmonized core. This is adoption by firm size, not use depth.

Observation verification: 2026-08-21. Edition date is not a new measurement date.

Sources and observation periods

To learn what changed in the work, change the evidence. The next two studies measure tasks and outputs directly. Their contrasting results are useful precisely because each is about a particular workflow.

Observed: different work, different measured effects.

Separate outcomes and populations; no combined AI-productivity average. Study-specific sources and exact-use review status below.

Two settings, two outcomes. These percentages have different denominators and are never pooled. Source-content checks are AI-assisted. Exact uses await author review.

Customer support

+15%

more issues resolved per hour

Higher throughput in this deployment

In one customer-support deployment, access to a GPT-3-based assistant increased issues resolved per hour by about 15%.

Tools: GPT-3-based assistant.

Period: Main rollout: fall 2020 / winter 2021.

One firm and an older assistant. Comparable counterfactual trends are an identification assumption. No measured national productivity, wage or welfare effect.

Who, comparison, uncertainty and quality

Population

5,172 agents at one business-software firm and subcontractors; most worked in the Philippines serving US customers.

Comparison

Assistant access versus no access in a staggered rollout

Design

Staggered rollout; difference-in-differences with agent, month and tenure controls. Not a randomized trial.

Uncertainty

Table II, column 3: +0.301 resolutions/hour; agent-clustered standard error 0.0329; 12,295 agent-month observations. The text reports 15.2%, rounded to 15% in the abstract.

Quality

Effects vary by skill. The highest-skilled agents show small quality declines; resolution data cover a subset because subcontractor quality records are incomplete.

QJE: Generative AI at Work — source and limits

Experienced software developers

+19%

longer task-completion time

Slower task completion in this experiment

METR's early-2025 experiment found 19% longer completion times in this selected setting; its February 2026 update warns that selection prevents a reliable estimate for later tools.

Tools: Primarily Cursor Pro with Claude 3.5 / 3.7 Sonnet.

Period: February–June 2025; methodology update 24 February 2026.

Selected developers and familiar repositories, using early-2025 tools. Later participation and task selection changed: people and tasks expected to benefit increasingly avoided AI-disallowed assignment. This is not an estimate of current models.

Who, comparison, uncertainty and quality

Population

16 experienced developers; 246 real tasks in mature open-source repositories they knew well.

Comparison

Task-level random assignment: AI allowed versus AI disallowed

Design

Task-level randomized experiment after issues were specified. Log-linear estimation controls for preregistered difficulty forecasts.

Uncertainty

Historical estimate: +19% task time; 95% interval +2% to +39%, as restated in the February 2026 update. Self-reported time supported by screen recordings; some post-review time imputed.

Quality

No demonstrated qualitative reduction in submitted PR quality. Post-review time difference was statistically insignificant.

METR: original experiment and 2026 update — sources and limits

Evidence and caveats

Both exact uses await the finite author/release review. Primary contents checked 6 September 2026 by Codex; no original replication or named specialist review. [Full evidence trace](#)

Where does a faster task stop short of a finished outcome?

A design might arrive sooner while its test still takes just as long. Or AI could improve the test as well. The interaction makes both possibilities available—and separates completing one project from sustaining repeated output.

Scenario: change the workflow assumptions.

Illustrative arithmetic, not an empirical model. Change a complement, expand where AI helps, or alter the dependencies.

Hypothetical inputs throughout. Days and accepted units/week are invented teaching units. No input comes from a benchmark, S mark or country estimate. The two examples below are independent models.

One project: when is it finished?

Design takes 8 days; testing takes 32. Change the speed of either stage, or change how they depend on each other.



Design 2 days · testing 32 days

Fixed scale: 40 days across

40 → 34 days to complete one project.

6 days saved · 15% less elapsed time · 1.18× project speed.

$$8/4 + 32/1 = 34 \text{ days}$$

Parallel means testing can proceed independently; it is a different process, not permission to test a design that does not yet exist. This model omits queues, rework and quality changes.

Change the baseline durations

Baseline design 8 days

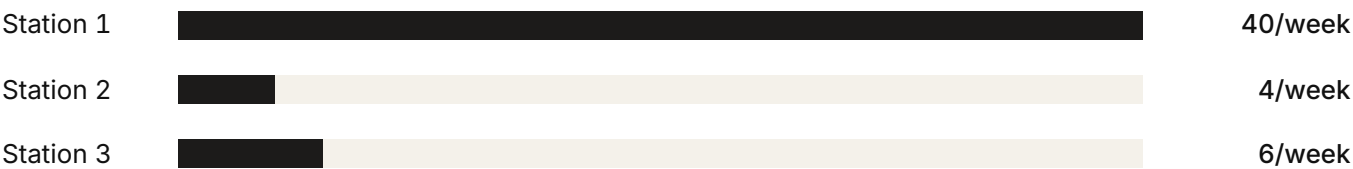


Baseline testing 32 days



Repeated work: how much can keep coming out?

Three stations perform different steps on the same accepted unit. Dedicated stations can work on different units at once; the slowest capacity limits steady output.

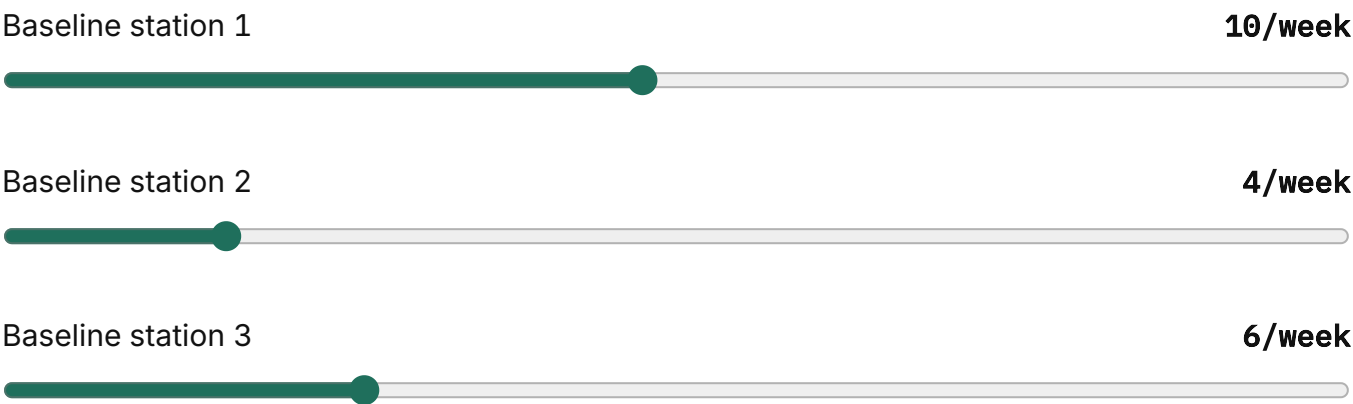


4 → 4 accepted units/week sustainable output ceiling.

$\min(40, 4, 6) = 4 \text{ units/week}$

Compatible units; adequate demand and buffers; no losses, rework, variability or start-up delay. Capacities are independently assumed, not inferred from the project durations. The bars rescale; read their labels.

Change baseline station capacities



A faster model can matter a great deal if it improves a limiting stage or changes the workflow. Speeding a stage already ahead of the others can have a smaller effect. Which case applies is an empirical question.

Hypothetical inputs adapted from the supplied author/editorial kit; equations and omissions shown at point of use.

SCENARIO

Which additional change would matter?

Fusion makes this question concrete. AI has already controlled plasma in physical hardware. The next question is what that result establishes, which conditions permit transfer, and how an improvement could reach an integrated plant outcome.

Observed: a real control result, with a longer evidence trail.

One research-tokamak demonstration and the additional evidence a plant-level conclusion would require. The arrows imply no measured timing or universal sequence.

Draft for release review · author reading pending. Degraeve et al., Nature (2022): a physical research/control demonstration. No plant effect size is estimated.

A demonstrated task

Observation

A controller trained in simulation operated TCV's physical plasma-control hardware. This is a result inside a real research tokamak.

Evidence to seek: Show robust performance across the operating conditions and machines relevant to the proposed next use.

A usable plant component

Interpretation: transfer requirement

A successful control demonstration does not itself qualify materials, plasma-facing components, blankets or fuel-cycle systems. Each has its own test and acceptance requirements.

Evidence to seek: Provide material-specific exposure and post-irradiation evidence matched to an acceptance standard. Planned irradiation is not a completed accepted dataset.

A useful plant outcome

Interpretation: wider evidence requirement

A plant-level conclusion would need integrated performance, maintainability, availability and an attributable effect on schedule or output. No date for fusion follows from this trace.

Evidence to seek: Observe integrated performance over an appropriate period and compare the outcome with a defensible counterfactual. Evidence outside this bank could change the conclusion.

A facility milestone is not evidence that AI caused that milestone. An absent result in this evidence bank is not proof of universal absence. Planned irradiation is a planned exposure requirement, not completed accepted qualification data. AI may change additional physical stages or the route between them; that possibility must be tested, not ruled out by the diagram. Degrave et al. — source, task and transfer limits

Degrave et al., Magnetic control of tokamak plasmas through deep reinforcement learning, Nature (2022). Draft use · author review pending

Follow the gain to its beneficiary.

Start with actors, then widen the geography. The customer-support study connects workers mainly in the Philippines with US customers. The fusion case connects a learned controller with a Swiss research facility. To reason about national capacity, one would also need evidence on access, replication, domestic value capture, distribution and adjustment costs. These cases supply no matched China-US causal comparison.

A stronger model could relax a quality threshold, improve a limiting stage, or make a different process viable. Another complement could be decisive instead. The contribution here is a way to make that question specific enough to investigate.

Why I started this — draft author introduction

Draft personal copy · for Jinhua's final edit

This project began with a question that sounded simple until I tried to answer it: when a country gets access to better AI, what does it actually get?

Conversations in Washington and my experience at the World AI Conference in Shanghai kept bringing me back to what happened next: in a workplace, a laboratory or a public institution. A better benchmark score seemed a useful beginning. It was a rather incomplete itinerary.

I want to follow the work far enough to ask who can use the tool, who does the work, and who receives the gain. A model provider, a worker and a country are different actors. A benefit to one need not measure the benefit to all three.

Source references

Observation periods remain source-specific. These original sources support the figures; exact caveats and review states accompany the figures and the evidence page.

Survey on the Access to Finance of Enterprises in the euro area - Fourth quarter of 2025 ·

European Central Bank · 2026-02-02

https://www.ecb.europa.eu/stats/ecb_surveys/safe/html/ecb.safe202602.en.html

Use of artificial intelligence in enterprises · Eurostat · 2026-06-02

[https://ec.europa.eu/eurostat/statistics-explained/index.php?](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises)

[title=Use_of_artificial_intelligence_in_enterprises](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises)

The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks ·

Kathryn Bonney; Cory Breau; Emin Dinlersoz; Lucia Foster; John Haltiwanger; Aditya Pande ·

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<https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-25.html>

Magnetic control of tokamak plasmas through deep reinforcement learning · Degraeve et al.;

Nature · 2022-02-16

<https://doi.org/10.1038/s41586-021-04301-9>

Generative AI at Work · Brynjolfsson; Li; Raymond · 2025-02-04

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(v2) · METR · 2025-07-25

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<https://metr.org/blog/2026-02-24-uplift-update/>